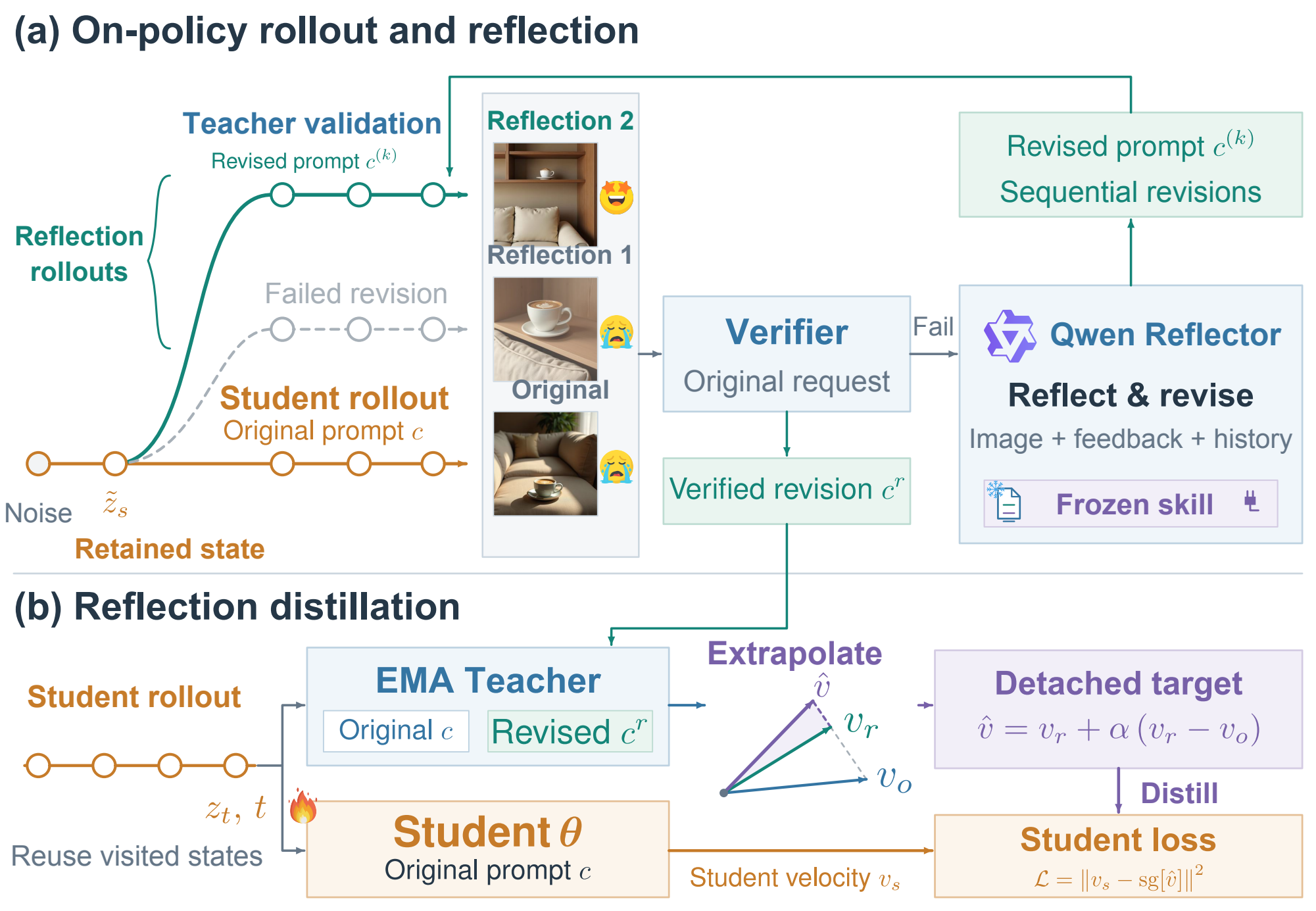

(a) On-policy rollout and reflection
Teacher validation
Revised prompt $c^{(k)}$
Reflection rollouts
Failed revision
Student rollout
Original prompt c
Noise
$\tilde{z}_s$
Retained state
Reflection 2
Reflection 1
Original
Verifier
Original request
Fail
Verified revision c^r
Qwen Reflector
Reflect & revise
Image + feedback + history
Frozen skill
Revised prompt $c^{(k)}$
Sequential revisions
(b) Reflection distillation
Student rollout
z_t, t
Reuse visited states
EMA Teacher
Original c
Revised c^r
Student θ
Original prompt c
Extrapolate
$\hat{v}$
v_r
v_o
Detached target
$\hat{v} = v_r + \alpha\,(v_r - v_o)$
Distill
Student loss
$\mathcal{L} = \|v_s - \mathrm{sg}[\hat{v}]\|^2$
Student velocity v_s